Functional Requirements

1. Customer Management

- Register new customers.
- View and update customer details.
- Delete customer records.
- Search customers by name, city, phone number, or email.

2. Product Management

- Add, update, view, and delete products.
- Search products by product ID or name.
- Filter products by name, price, price range, and company.
- View product details and available stock.

3. Customer Login

- Allow customers to register and log in.
- Allow customers to log out securely.
- Restrict shopping and ordering to logged-in customers.

4. Shopping Cart

- Add products to the cart.
- Update product quantities.
- Remove products from the cart.
- View cart items and total amount.
- Check product availability before checkout.

5. Order Management

- Place orders after customer confirmation.
- Store and retrieve customer orders.
- Calculate product subtotal, shipping cost, discounts, and final amount.
- Update order status.
- View order history and tracking details.

6. Payment Management

- Support UPI payments.
- Support debit and credit card payments.
- Support Cash on Delivery (COD).
- Support gift card payments.

7. Order Tracking

- Track dispatched orders.
- Display courier and transit status.
- Update orders as delivered or returned.

8. Subscription Management

- Provide monthly, quarterly, half-yearly, and yearly subscriptions.
- Offer free shipping to active subscribers.

9. Discount Management

- Provide discount coupons based on qualifying order history.
- Check coupon eligibility.

- Apply eligible discounts to orders.

The OOP concepts demonstrated are

1. Encapsulation and data hiding

Example: Customer, Product, CartItem, and Order.

- Keep fields private.

- Validate updates through public methods or constructors.

- Don't expose a mutable internal collection directly.

- Don't let callers set an arbitrary order status without checking the transition.

```
public class CartItem {
    private final Product product;
    private int quantity;
    public CartItem(Product product,int quantity) {
        if(product==null||quantity<=0) {
            throw new IllegalArgumentException("Invalid cart item");
        }
        this.product=product;
        this.quantity=quantity;
    }

    public void changeQuantity(int quantity) {
        if(quantity<=0) {
            throw new IllegalArgumentException("Quantity must be
positive");
        }
        this.quantity=quantity;
```

```
    }

    public int getQuantity() {
        return quantity;
    }
}
```

2. Abstraction

Example: PaymentMethod and ShippingPolicy.

A checkout service should request a payment without needing to know the internal details of every payment type.

```
public interface PaymentMethod {
    PaymentResult pay(BigDecimal amount);
}
```

CheckoutService depends on this contract, not on the internal implementation of UPI, cards, COD, or gift cards.

3. Inheritance

Example: customer account types or payment types.

We could create an abstract CustomerAccount with common account behavior, then derive RegularCustomerAccount and SubscribedCustomerAccount.

A subscription is often better represented as a separate Subscription object associated with a Customer, because it has its own plan, expiry, and lifecycle.

Payment implementations are a natural place to demonstrate inheritance through an abstract payment class, with an interface defining the contract.

4. Polymorphism

Example: one checkout workflow supporting multiple payment methods.

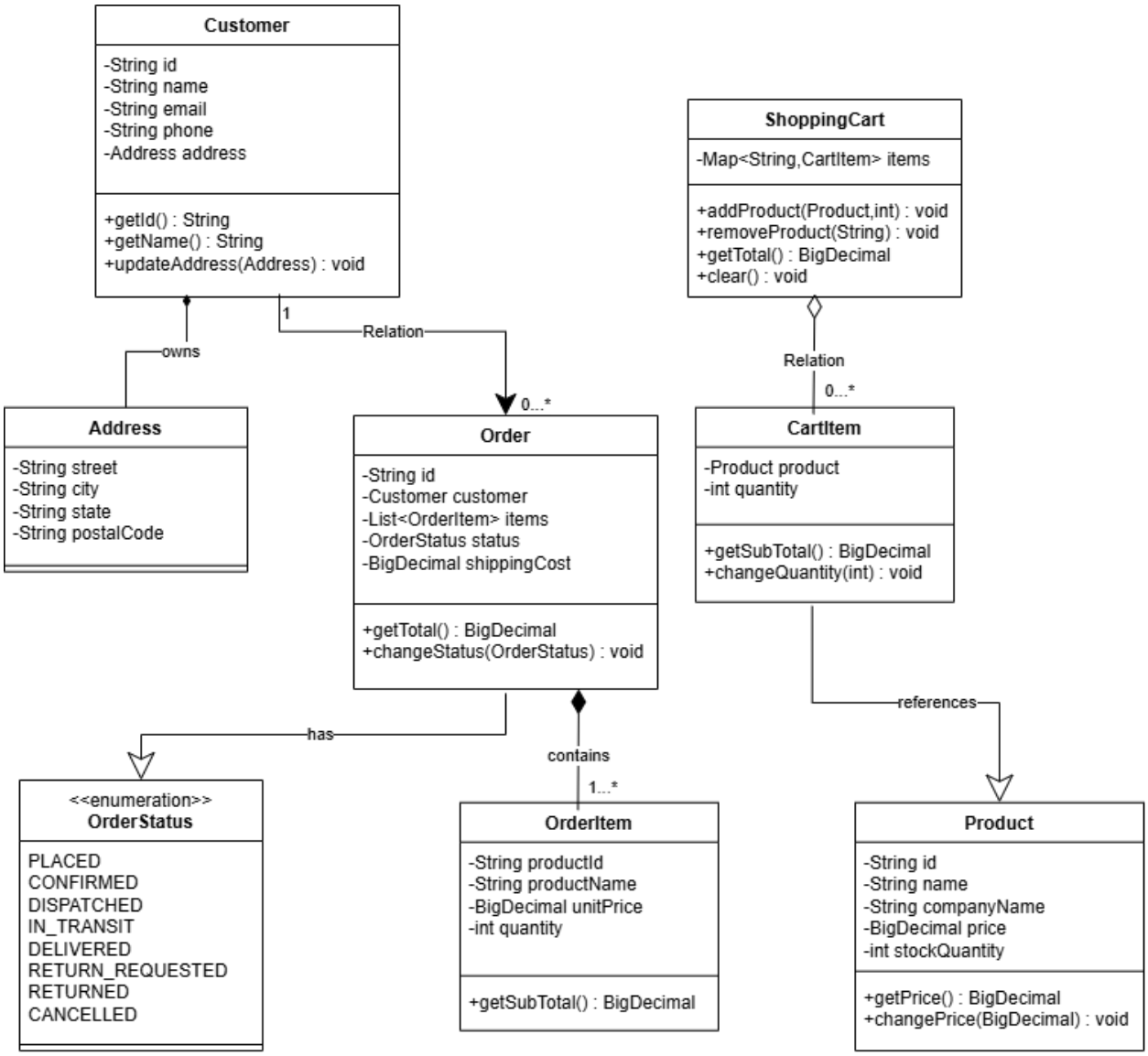
```
PaymentMethod paymentMethod = new UpiPayment();

PaymentResult result = paymentMethod.pay(amount);
```

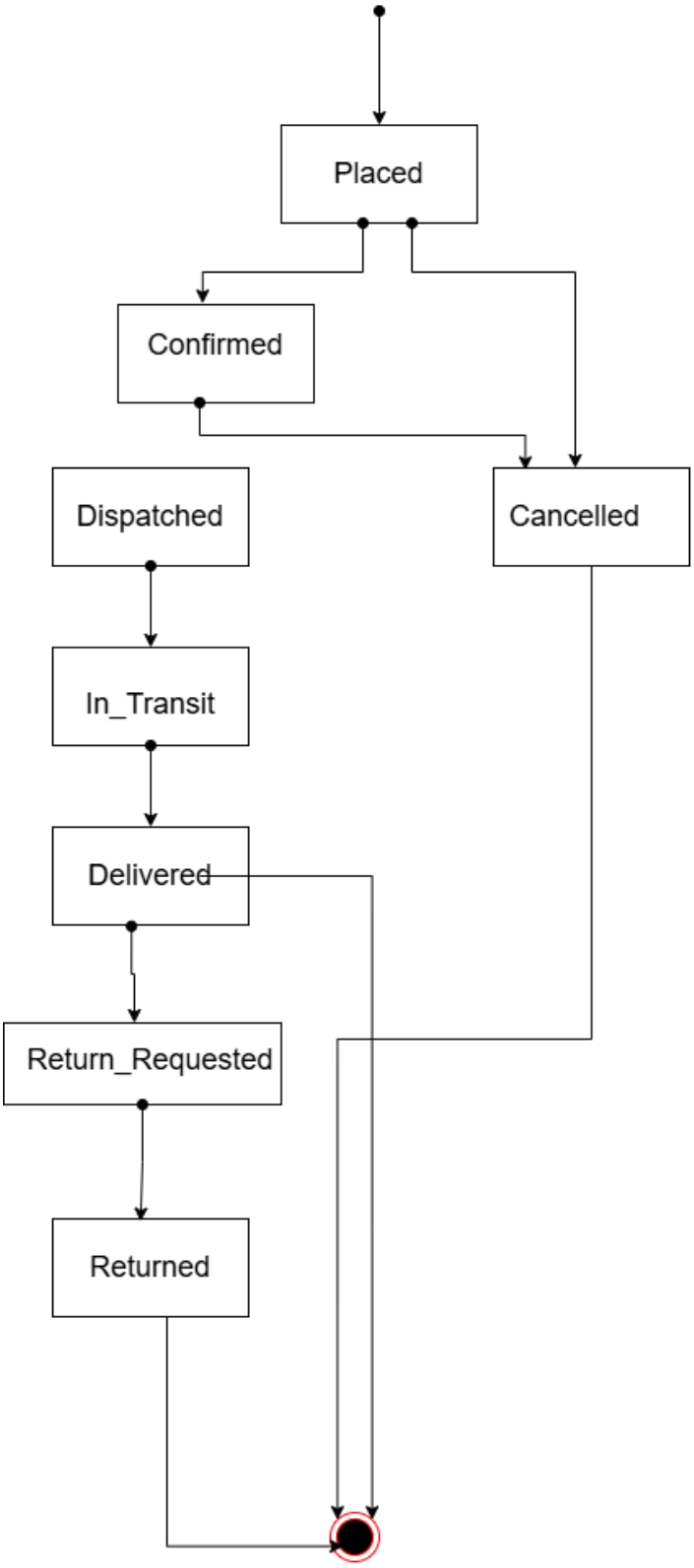
The exact same checkout service can receive CardPayment, CashOnDeliveryPayment, or GiftCardPayment.]

Alos demonstrated composition over inheritance, constructor injection, immutability where appropriate, and the Single Responsibility Principle.

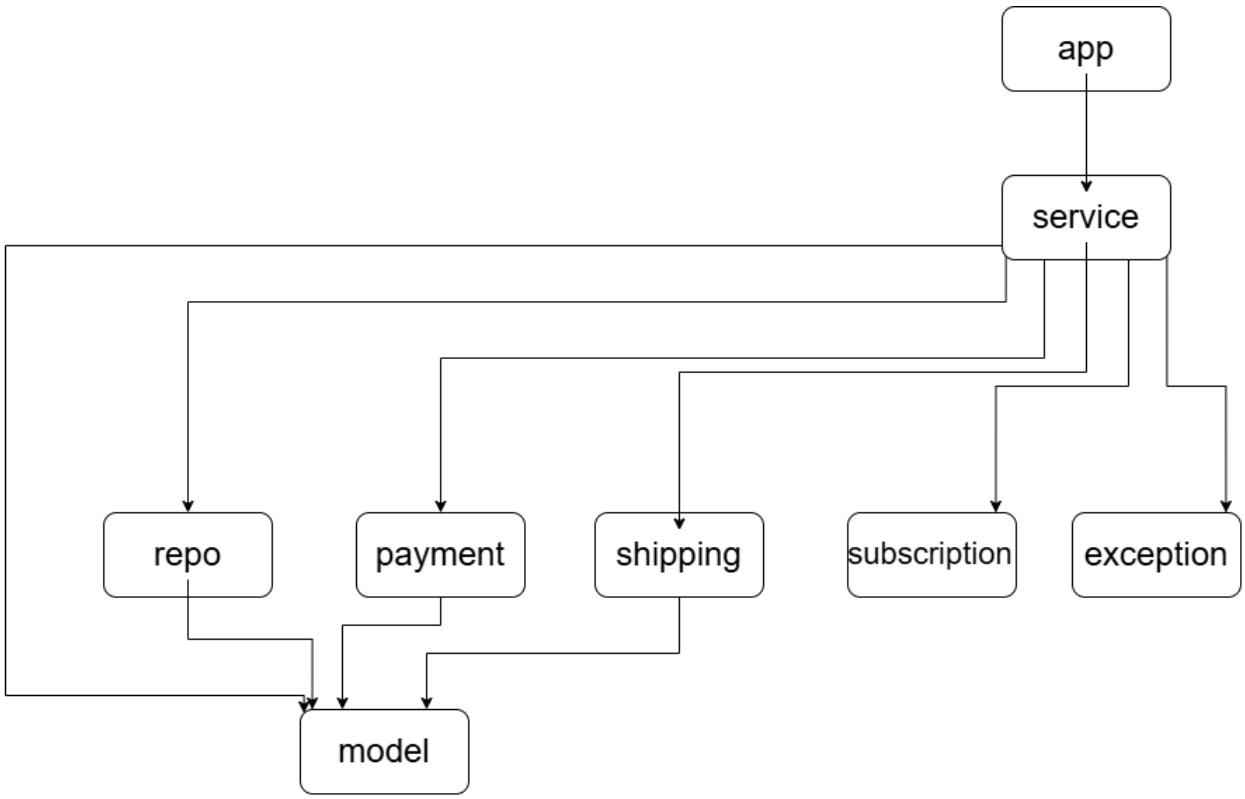
Class Diagrams



Order State


Placed
Confirmed
Dispatched
Cancelled
In_Transit
Delivered
Return_Requested
Returned

Package Dependency



Implementation

01

Domain models and validation

Customer, Address, Product, OrderItem, CartItem, and enums. Use private fields, constructors, validation, and appropriate immutable fields.

Deliverable: valid objects that cannot easily enter invalid states.

02

Repositories and search

Customer and product CRUD operations, email and phone lookup, city filtering, product-name matching, company filtering, exact prices, and price ranges.

Deliverable: testable in-memory data management.

03

Services and authentication

Registration, login, customer lookup, duplicate-email prevention, and admin-only operations.

Deliverable: business rules separate from storage.

04

Shopping cart and checkout

Add and remove items, update quantities, check stock, calculate subtotal, calculate shipping, and generate orders.

Deliverable: end-to-end purchase flow.

05

Order management and tracking

Valid state transitions, admin updates, timestamps, courier information, tracking events, and returns.

Deliverable: a realistic order lifecycle.

06

Payment, subscription, and coupons

Payment polymorphism, gift-card balances, subscription expiry, free shipping, and eligibility-based discounts.

Deliverable: advanced OOP business rules.

Functional Requirements

(implementation w.r.t each functional requirement)

1. Customer Management

- Register new customers.

```
Customer registration
Name: swamy
Email: swamy@gmail.com
Phone: 9101021321
Create password (minimum 6 characters): 123456
Street: Day Oak
City: Gudem
State: Assam
Postal code: 210220
Country: India
Registration complete. You can now log in with swamy@gmail.com.
```

-

View and update customer details.

```
Choose an option: 5
Search customers: 1. Name  2. City  3. Phone  4. Email
Choose a search type: akhil
Enter a whole number.
Choose a search type: 1
Name contains: akhil
ID 48fe0a51-5267-42e1-bd64-dc6158113c8c | akhil | akhi@g.com | city Pole
```

- Delete customer records.

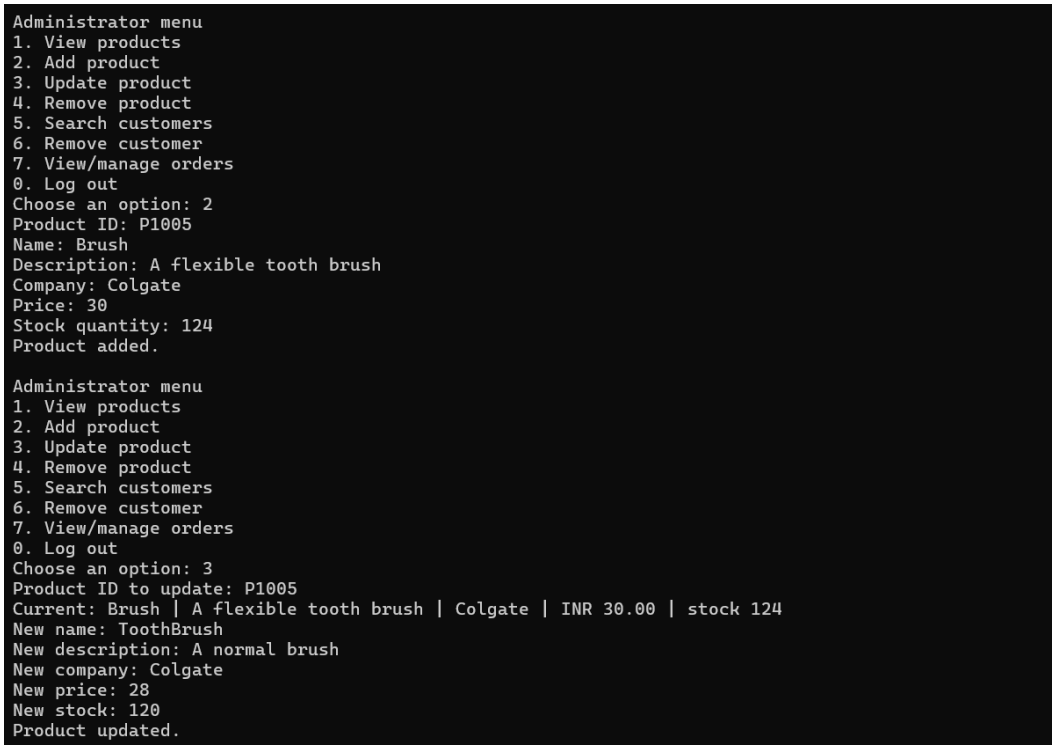
```
Administrator menu
1. View products
2. Add product
3. Update product
4. Remove product
5. Search customers
6. Remove customer
7. View/manage orders
0. Log out
Choose an option: 6
Customer ID to remove: a6afe5c8-6696-404c-bae2-690ff8d86929
Customer account removed.
```

- Search customers by name, city, phone number, or email.

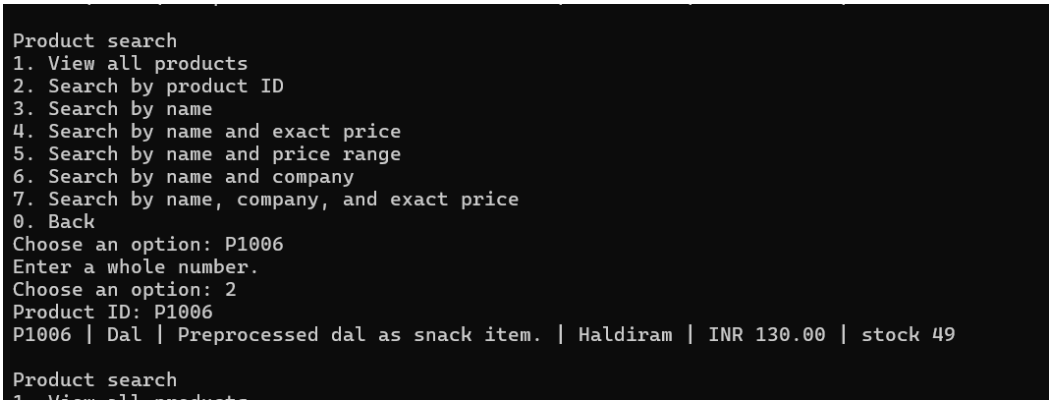
```
0. Log out
Choose an option: 5
Search customers: 1. Name  2. City  3. Phone  4. Email
Choose a search type: 4
Email: akhi@g.com
ID 48fe0a51-5267-42e1-bd64-dc6158113c8c | akhil | akhi@g.com | city Pole
```

2. Product Management

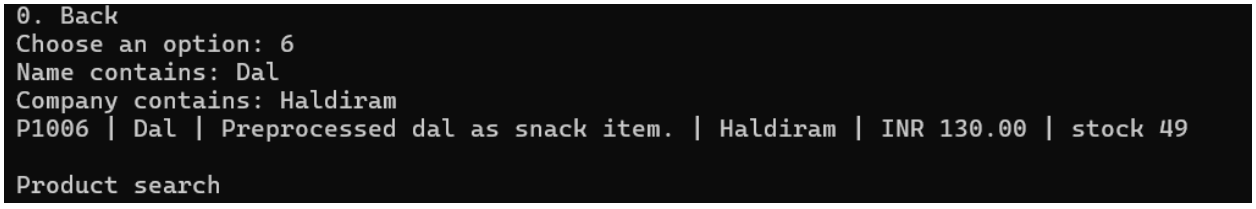
- Add, update, view, and delete products.



- Search products by product ID or name.



- Filter products by name, price, price range, and company.



```
Product search
1. View all products
2. Search by product ID
3. Search by name
4. Search by name and exact price
5. Search by name and price range
6. Search by name and company
7. Search by name, company, and exact price
0. Back
Choose an option: 5
Name contains: dal
Minimum price: 10
Maximum price: 150
P1002 | Toor Dal | Split pigeon peas | Farm Fresh | INR 95.00 | stock 25
P1006 | Dal | Preprocessed dal as snack item. | Haldiram | INR 130.00 | stock 49

Product search
```

- View product details and available stock.

```
Choose an option: 1
P1001 | Basmati Rice | Premium long-grain rice | KiloKart Select | INR 120.00 | stock 30
P1002 | Toor Dal | Split pigeon peas | Farm Fresh | INR 95.00 | stock 25
P1003 | Olive Oil | Extra virgin cooking oil | Oliva | INR 450.00 | stock 12
P1004 | Green Tea | Organic green tea bags | Leaf & Cup | INR 180.00 | stock 18
P1005 | ToothBrush | A normal brush | Colgate | INR 28.00 | stock 120
P1006 | Dal | Preprocessed dal as snack item. | Haldiram | INR 130.00 | stock 49

Product search
```

3. Customer Login

- Allow customers to register and log in.

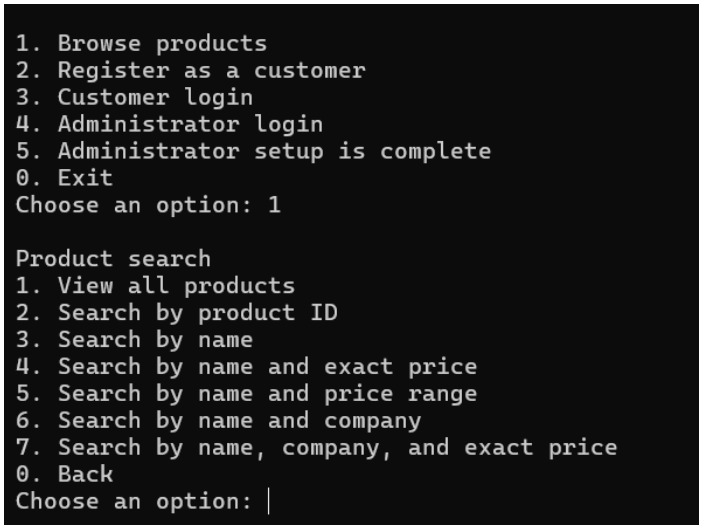
```
Customer registration
Name: surya
Email: surya@g.com
Phone: 123
Create password (minimum 6 characters): 123456
Street: Ion
City: Negra
State: PL
Postal code: 901021
Country: India
Registration complete. You can now log in with surya@g.com.

1. Browse products
2. Register as a customer
3. Customer login
4. Administrator login
5. Administrator setup is complete
0. Exit
Choose an option: 3
Email: surya@g.com
Password: 123456
Welcome, surya.
```

- Allow customers to log out securely.

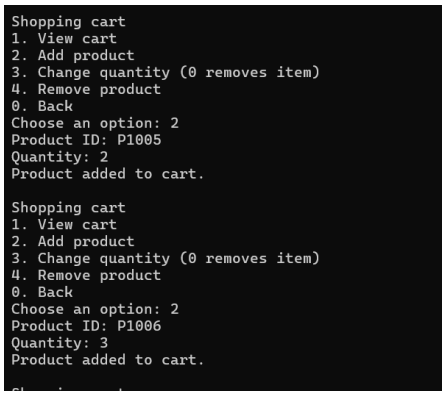
```
Customer menu - surya
1. Browse/search products
2. Manage shopping cart
3. Checkout
4. View my orders and tracking
5. Update my profile
6. Manage subscription
0. Log out
Choose an option: 0
You have been logged out.
```

- Restrict shopping and ordering to logged-in customers.

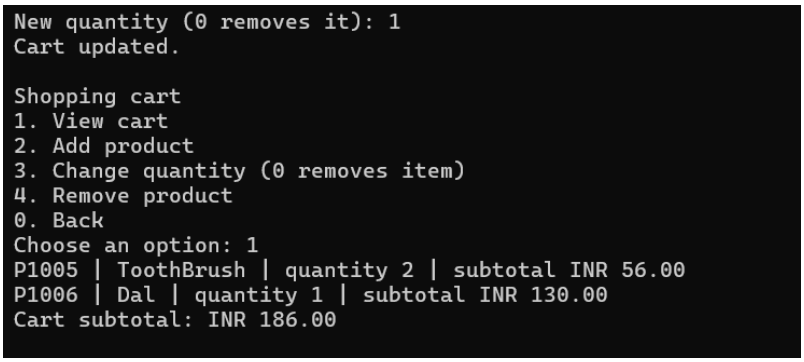


4. Shopping Cart

- Add products to the cart.

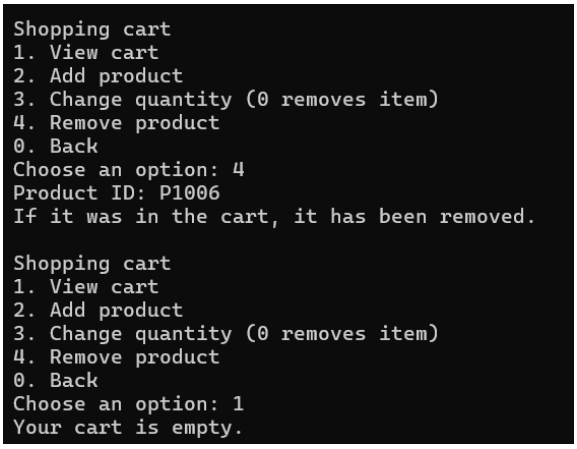


- Update product quantities.

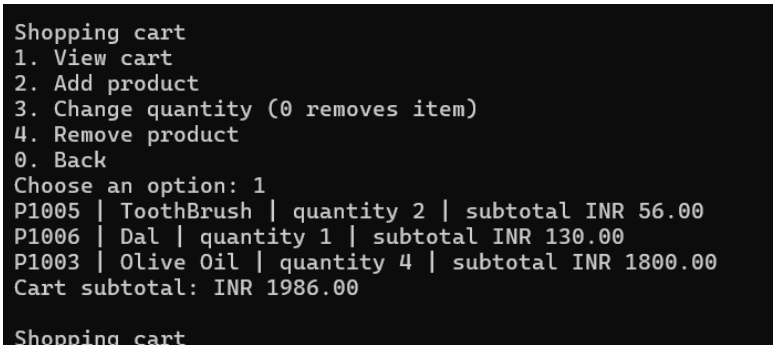


- Remove products from the cart.



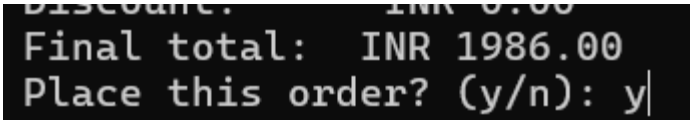


- View cart items and total amount.

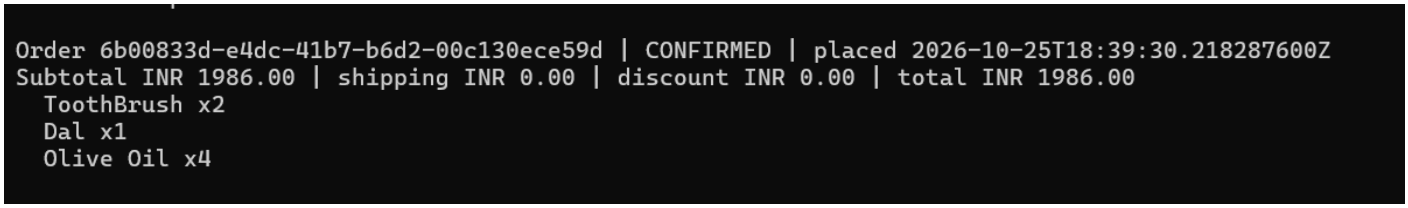


5. Order Management

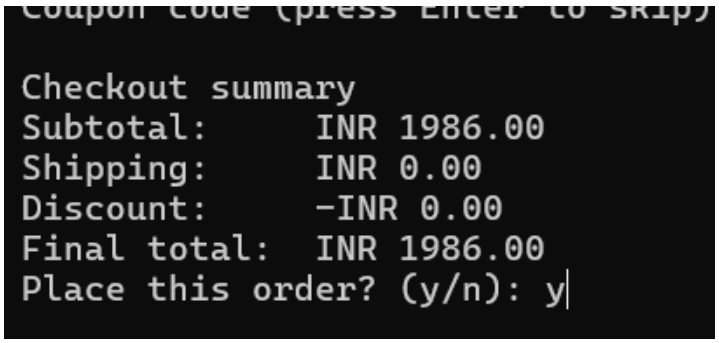
- Place orders after customer confirmation.



- Store and retrieve customer orders.



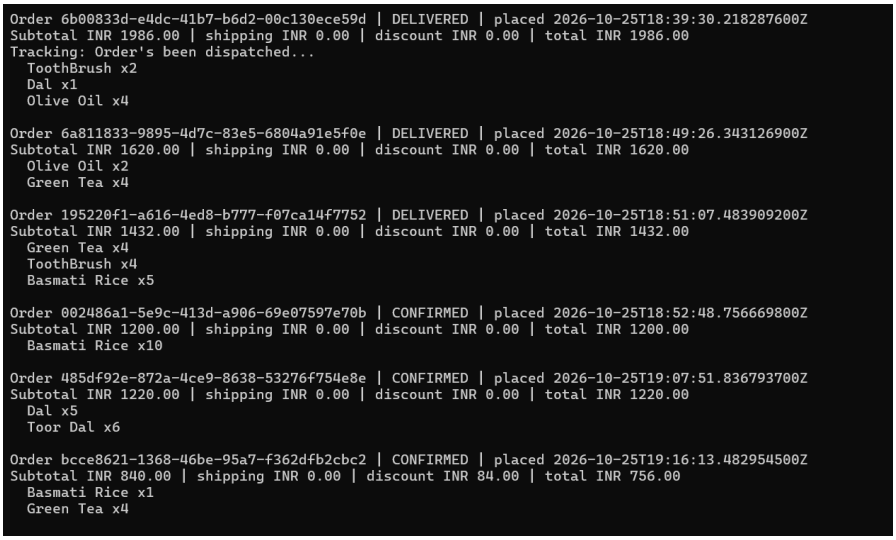
- Calculate product subtotal, shipping cost, discounts, and final amount.



- Update order status.

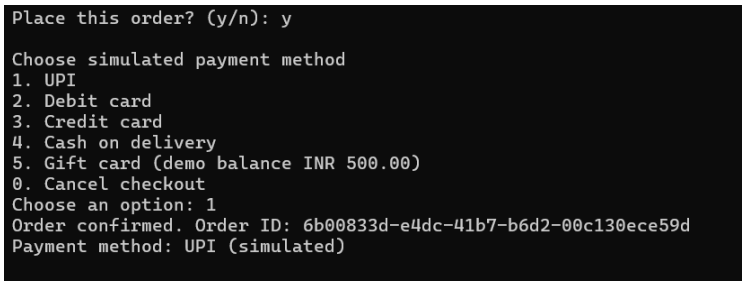


- View order history and tracking details.

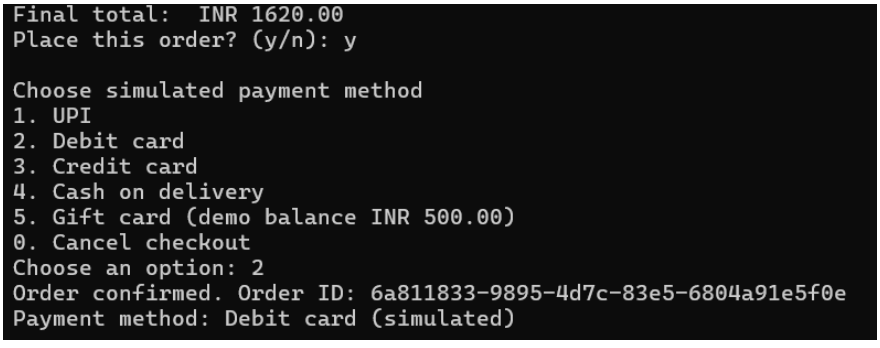


6. Payment Management

- Support UPI payments.



- Support debit and credit card payments.



- Support Cash on Delivery (COD).

```
Final total:  INR 1432.00
Place this order? (y/n): y

Choose simulated payment method
1. UPI
2. Debit card
3. Credit card
4. Cash on delivery
5. Gift card (demo balance INR 500.00)
0. Cancel checkout
Choose an option: 4
Order confirmed. Order ID: 195220f1-a616-4ed8-b777-f07ca14f7752
Payment method: Cash on delivery

Customer menu — surya
```

- Support gift card payments.

```
Place this order? (y/n): y

Choose simulated payment method
1. UPI
2. Debit card
3. Credit card
4. Cash on delivery
5. Gift card (demo balance INR 500.00)
0. Cancel checkout
Choose an option: 5
Could not complete that action: Payment was declined or the gift-card balance was insufficient
```

7. Order Tracking

- Track dispatched orders.

```
4. View my orders and tracking
5. Update my profile
6. Manage subscription
0. Log out
Choose an option: 4

Order 6b00833d-e4dc-41b7-b6d2-00c130ece59d | DELIVERED | placed 2026-10-25T18:39:30.218287600Z
Subtotal INR 1986.00 | shipping INR 0.00 | discount INR 0.00 | total INR 1986.00
Tracking: Order's been dispatched...
  ToothBrush x2
  Dal x1
  Olive Oil x4

Order 6a811833-9895-4d7c-83e5-6804a91e5f0e | CONFIRMED | placed 2026-10-25T18:49:26.343126900Z
Subtotal INR 1620.00 | shipping INR 0.00 | discount INR 0.00 | total INR 1620.00
  Olive Oil x2
  Green Tea x4
```

- Display courier and transit status.

```
Update an order status? (y/n): y
Order ID: 6a811833-9895-4d7c-83e5-6804a91e5f0e
Current status: CONFIRMED
1. PLACED
2. CONFIRMED
3. DISPATCHED
4. IN_TRANSIT
5. DELIVERED
6. RETURN_REQUESTED
7. RETURNED
8. CANCELLED
New status: 3
Tracking information (blank to leave unchanged):
Order status updated to DISPATCHED.
```

- Update orders as delivered or returned.

```
Order 6b00833d-e4dc-41b7-b6d2-00c130ece59d | DELIVERED | placed 2026-10-25T18:39:30.218287600
Subtotal INR 1986.00 | shipping INR 0.00 | discount INR 0.00 | total INR 1986.00
```

8. Subscription Management

- Provide monthly, quarterly, half-yearly, and yearly subscriptions.

```
6. Manage subscription
0. Log out
Choose an option: 6
1. Monthly
2. Quarterly
3. Half-yearly
4. Yearly
0. Back
Choose a subscription duration: 3
Activate the HALF_YEARLY subscription? This is a local demo with no charge (y/n): y
Subscription active until 2027-04-26. You now receive free shipping.
```

- Offer free shipping to active subscribers.

```
Coupon code (press Enter to ski

Checkout summary
Subtotal:      INR 1220.00
Shipping:      INR 0.00
Discount:      -INR 0.00
Final total:   INR 1220.00
Place this order? (y/n): y
```

9. Discount Management

- Provide discount coupons based on qualifying order history.

```
Order 6b00833d-e4dc-41b7-b6d2-00c130ece59d | DELIVERED | placed 2026-10-25T18:39:30.218287600Z
Subtotal INR 1986.00 | shipping INR 0.00 | discount INR 0.00 | total INR 1986.00
Tracking: Order's been dispatched...
  ToothBrush x2
  Dal x1
  Olive Oil x4

Order 6a811833-9895-4d7c-83e5-6804a91e5f0e | DELIVERED | placed 2026-10-25T18:49:26.343126900Z
Subtotal INR 1620.00 | shipping INR 0.00 | discount INR 0.00 | total INR 1620.00
  Olive Oil x2
  Green Tea x4

Order 195220f1-a616-4ed8-b777-f07ca14f7752 | DELIVERED | placed 2026-10-25T18:51:07.483909200Z
Subtotal INR 1432.00 | shipping INR 0.00 | discount INR 0.00 | total INR 1432.00
  Green Tea x4
  ToothBrush x4
  Basmati Rice x5
```

- Check coupon eligibility.

```
0. Log out
Choose an option: 3
Coupon code (press Enter to skip): LOYAL10
Could not complete that action: Coupon requires 3 previous qualifying orders
```

- Apply eligible discounts to orders.

```
Customer menu - surya
1. Browse/search products
2. Manage shopping cart
3. Checkout
4. View my orders and tracking
5. Update my profile
6. Manage subscription
0. Log out
Choose an option: 3
Coupon code (press Enter to skip): LOYAL10

Checkout summary
Subtotal:      INR 840.00
Shipping:      INR 0.00
Discount:      -INR 84.00
Final total:   INR 756.00
```

Admin

```
First-run administrator setup. Choose your own credentials.
Name: bss
Email: bss@gmail.com
Phone: 9010112345
Create password (minimum 6 characters): 123456
Administrator account created. No password is displayed or stored as plain text.

1. Browse products
2. Register as a customer
3. Customer login
4. Administrator login
5. Administrator setup is complete
0. Exit
Choose an option: 4
Administrator email: bss@gmail.com
Password: 123456
Administrator access granted. Welcome, bss.
```